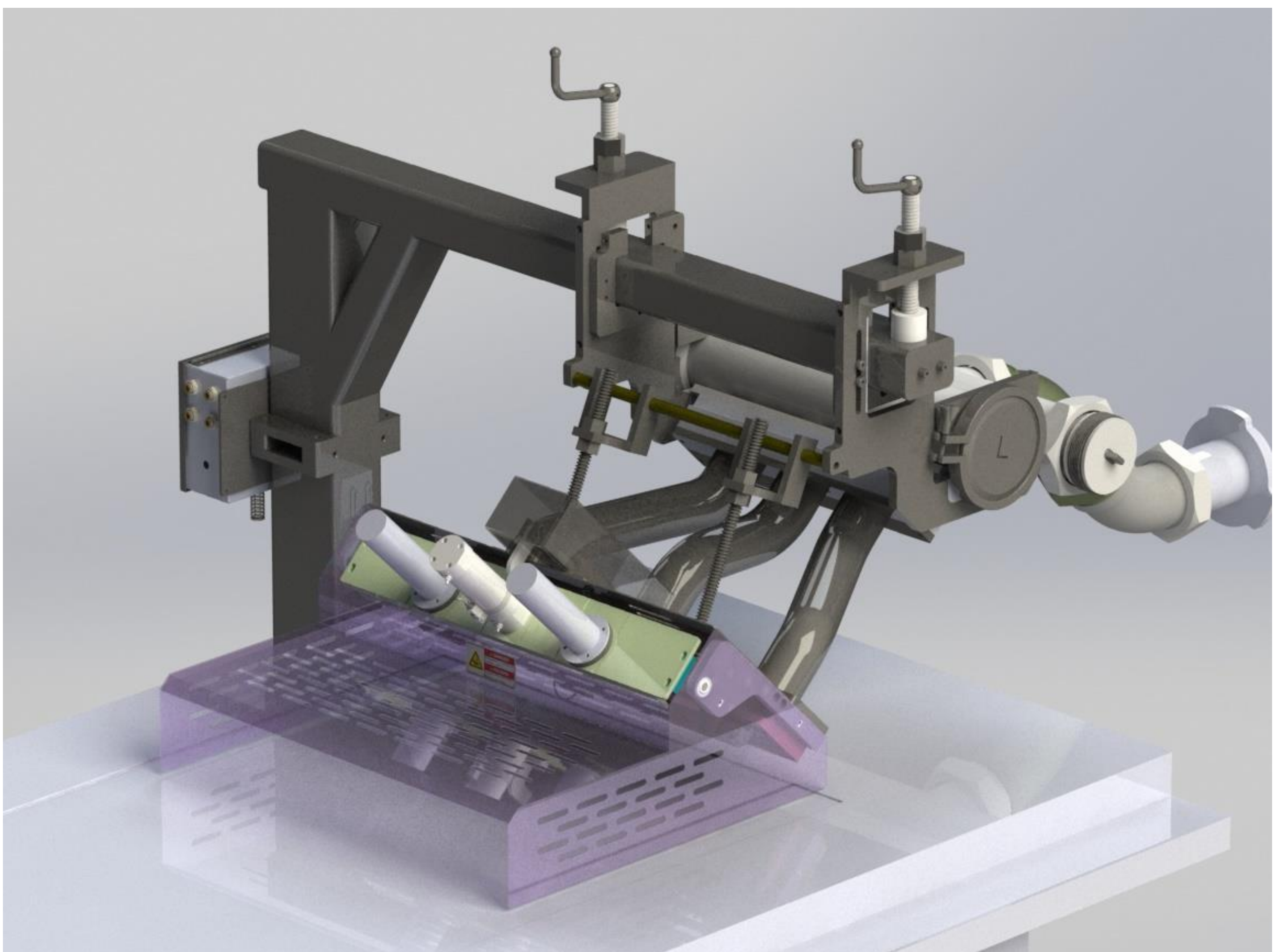
MULTI-LANE PNEUMATIC GUILLOTINE CUTOFF

Owners Manual

PART NUMBER: PR000469



DOC-1004301 REVISION DATE: 03/23/2017

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1. DESCRIPTION

The multi-lane guillotine cutoff is designed to attach to a Vemag filling machine to form and cut various food products into different shapes. A simple change of shaping block can dramatically alter the shape of the product being made, resulting in near endless possibilities.

The cutoff attaches to the outlets of the Reiser Waterwheel. A stainless steel manifold may be used to adapt from the centerlines of the Waterwheel to the final product centerlines and/or to transition to product sizes larger than the standard waterwheel outlet area if required.

Cutting is done by means of a stainless steel knife driven by a pneumatic air cylinder.

Each cutoff is custom designed to match customer specifications. Images in this manual may vary from your cutoff.

Reiser's engineering department can design custom shaping blocks to various shapes and sizes to meet your product needs.

If you have any questions related to the operation or service of this attachment, please contact Reiser customer service.

REISER
725 DEDHAM STREET
CANTON, MA 02021
(781) 821-1290

2. SAFETY INSTRUCTIONS

The Multilane Pneumatic Guillotine Cutoff:

- ❑ is supplied as an attachment for the Vemag V500, DP, HP, Robby, and Robby II machines.
- ❑ is used in conjunction with the Reiser Waterwheel flow divider.
- ❑ is designed and built for cutting and shaping food products.
- ❑ is intended to process raw materials at a temperature between -4°C and 50°C (25°F and 120°F).
- ❑ is designed to be mounted over a Reiser conveyor system.
- ❑ should only be operated by trained operators.
- ❑ should only be cleaned by trained cleaning staff.
- ❑ should only be serviced by trained maintenance staff.

2. SAFETY INSTRUCTIONS

ATTENTION – IMPORTANT SAFETY WARNINGS REGARDING THIS PRODUCT

AIR POWERED KNIFE CAN CAUSE SERIOUS INJURY INCLUDING AMPUTATION

THE FILLING MACHINE MUST BE TURNED OFF, ACCESSORY PLUG DISCONNECTED, AND AIR SUPPLY MUST ALWAYS BE DISCONNECTED DURING SETUP, CLEANING, AND SERVICING.

KNIFE COULD UNEXPECTEDLY MOVE

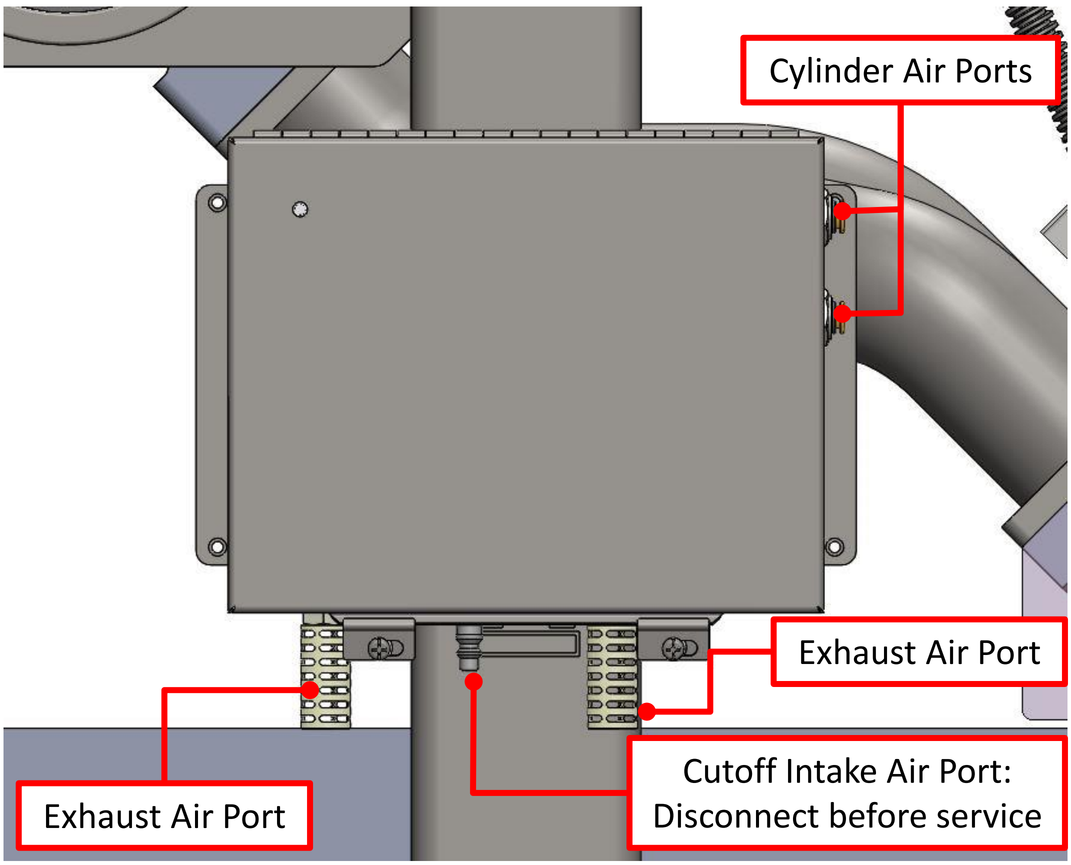
THE MULTILANE PNEUMATIC GUILLOTINE CUTOFF USES COMPRESSED AIR AS AN ENERGY SOURCE. EVEN IF THE KNIFE IS NOT MOVING, IT COULD SUDDENLY OPEN OR CLOSE CAUSING INJURY OR AMPUTATION. REMOVE ALL RESIDUAL ENERGY AND VERIFY A ZERO ENERGY STATE EXISTS.

LOCKOUT/TAGOUT

BE SURE TO FOLLOW ALL PROCEDURES FOR CONTROLLING ALL HAZARDOUS ENERGY SOURCES ESTABLISHED BY THE EMPLOYER.

2. SAFETY INSTRUCTIONS

1. **ALWAYS TURN OFF THE VEMAG FILLER AND DISCONNECT THE AIR TO THE CUTOFF VALVE BEFORE ATTEMPTING TO SETUP, SERVICE, OR CLEAN THE CUTOFF.**



2. **THE KNIFE EDGE(S) IS VERY SHARP AND CAN CAUSE SERIOUS CUTS. USE CAUTION AND CUT-PROOF PERSONAL PROTECTIVE EQUIPMENT WHEN HANDLING OR WORKING AROUND KNIFE ASSEMBLY.**

2. SAFETY INSTRUCTIONS

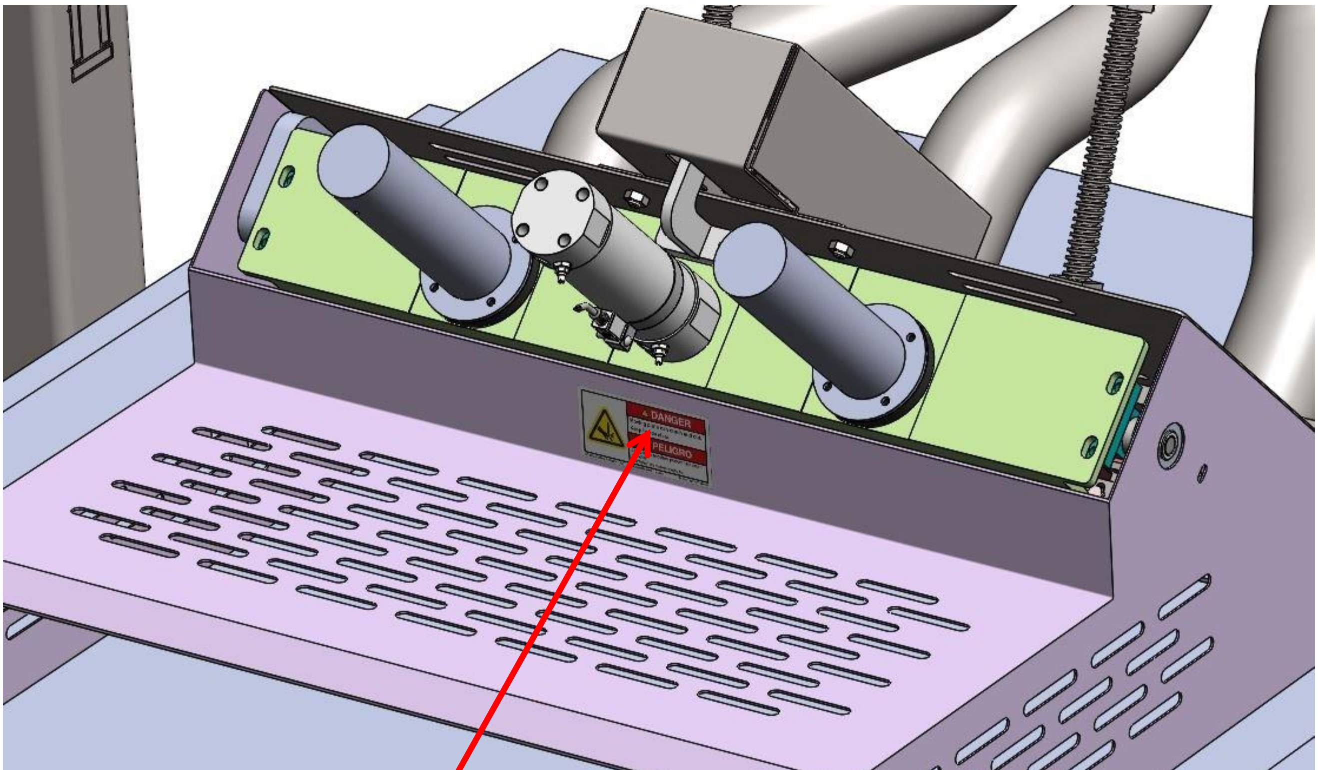
3. **NEVER OPERATE THE CUTOFF WITHOUT A CONVEYOR AND GUARDS SECURED IN PLACE.**

4. **THE CUTOFF MUST BE PROPERLY POSITIONED OVER THE CONVEYING SYSTEM WITH NO MORE THAN 3/8” (9.5MM) GAP BETWEEN THE GUARDING AND THE CONVEYOR.**

5. **ALWAYS ENSURE THAT THE STAND IS FASTENED TO THE CONVEYOR DOCK AND CUTOFF IS LOWERED INTO POSITION WITH JAM NUTS TIGHT BEFORE CONNECTING THE AIR SUPPLY, CONNECTING THE ACCESSORY CABLE TO THE FILLING MACHINE, AND TURNING THE MACHINE ON.**

2. SAFETY INSTRUCTIONS

6. **NEVER OPERATE THE CUTOFF WITHOUT THE CORRECT WARNING LABELS IN PLACE. IF YOU NEED REPLACEMENT LABELS, PLEASE CONTACT REISER CUSTOMER SERVICE.**



Warning Label
P/N MP-010-001

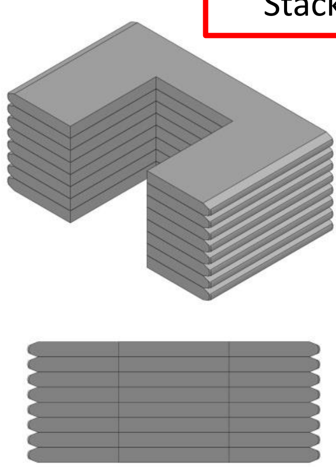
3. WATERWHEEL SETUP

Before assembling the Waterwheel, it is important to inspect the internal components for wear or damage. A worn drive hub, bent vane, or damaged drive rod will seriously affect the performance of the Waterwheel.

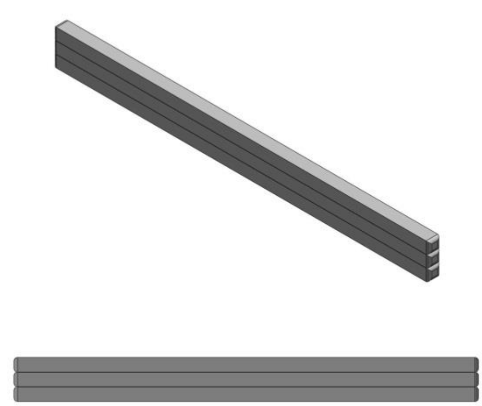
Visually inspect the chambers and hubs for grooves, dents, scratches, and nicks. If minor damage is detected, it can usually be remedied with some emery cloth. If the component cannot be repaired, discard the component and replace it with a new one.

On a flat surface, stack the vanes, one on top of another and inspect for bent vanes. Do the same with the square drive rods, and inspect for twisting. If any component is found to be defective, discard the component and replace it with a new one.

Stacked vanes

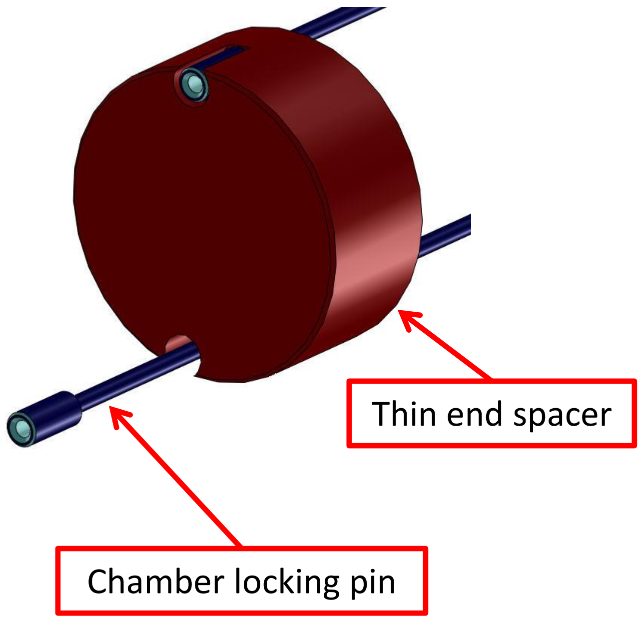


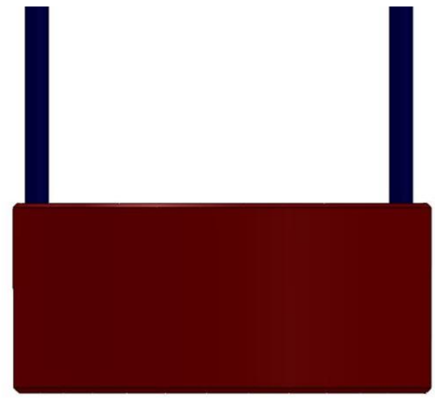
Stacked rods



3. WATERWHEEL SETUP

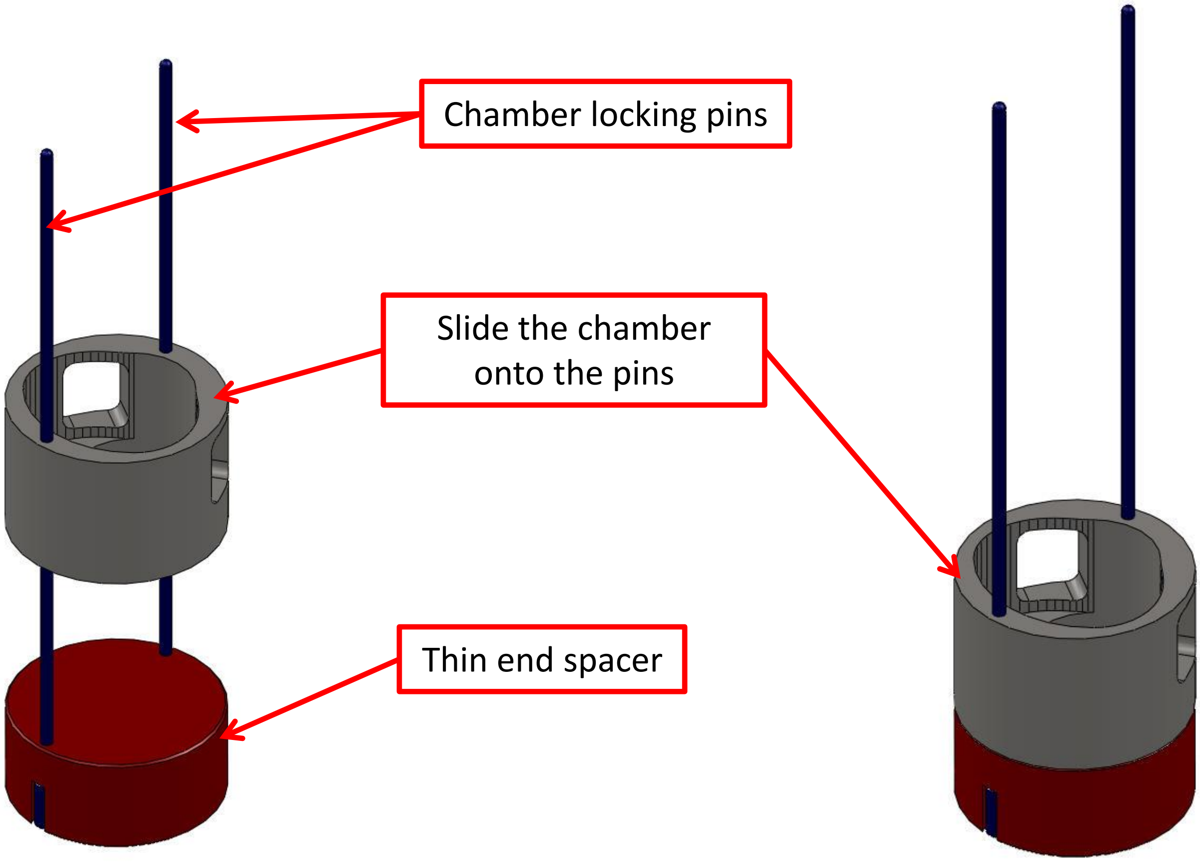
1. Insert the chamber locking pins into the **THIN** end spacer, and stand the assembly upright.



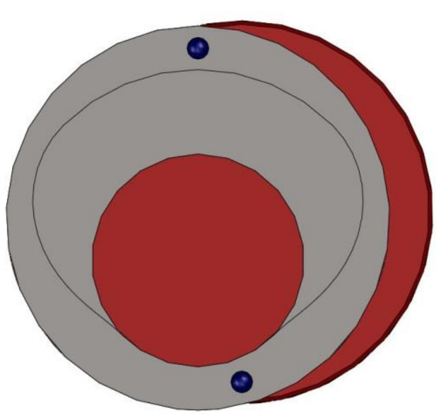


2. Slide a chamber over the locking pins with the open side of the chamber facing up. The chamber should be centered on the thin end spacer. If it is not, remove the chamber and reverse the holes onto the pins.

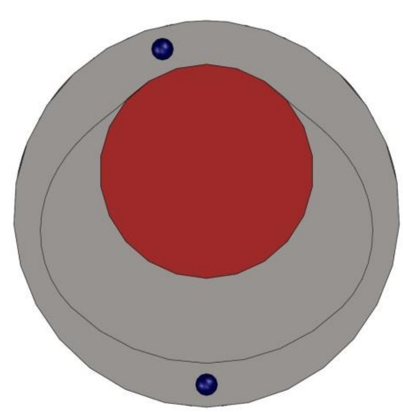
3. WATERWHEEL SETUP



<u>Incorrect</u> assembly; reverse the holes onto the pins

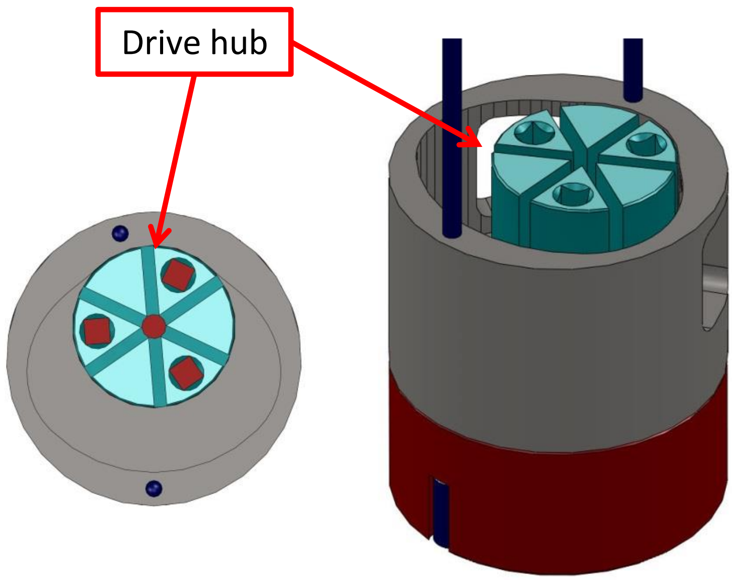


Correct assembly



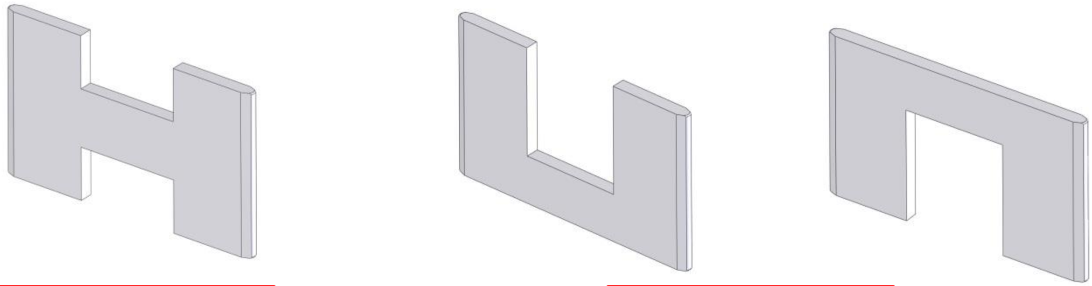
3. WATERWHEEL SETUP

3. Place a drive hub into the chamber with the slots facing upward.



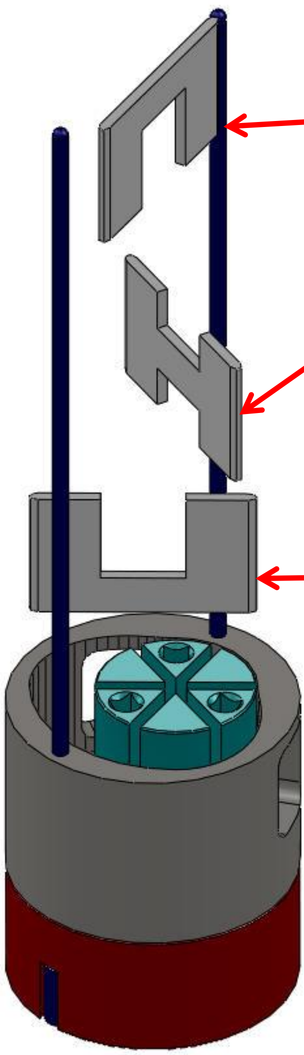
4. Install the vanes into the drive hub. The drive hub will require a total of three (3) vanes. Two of the vanes are shaped like a “U”, and one of them is shaped like an “H”. Install one of the “U” shaped vanes into one of the slots in the drive hub, with the notch facing upward. Next, install the “H” shaped vane into one of the slots in the drive hub. Lastly, install the remaining “U” shaped vane into the remaining slot in the drive hub, with the notch facing downward.

3. WATERWHEEL SETUP



“H” vane

“U” vanes

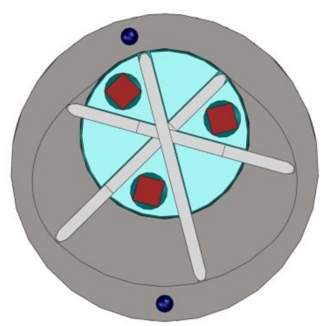


Install last

Install second

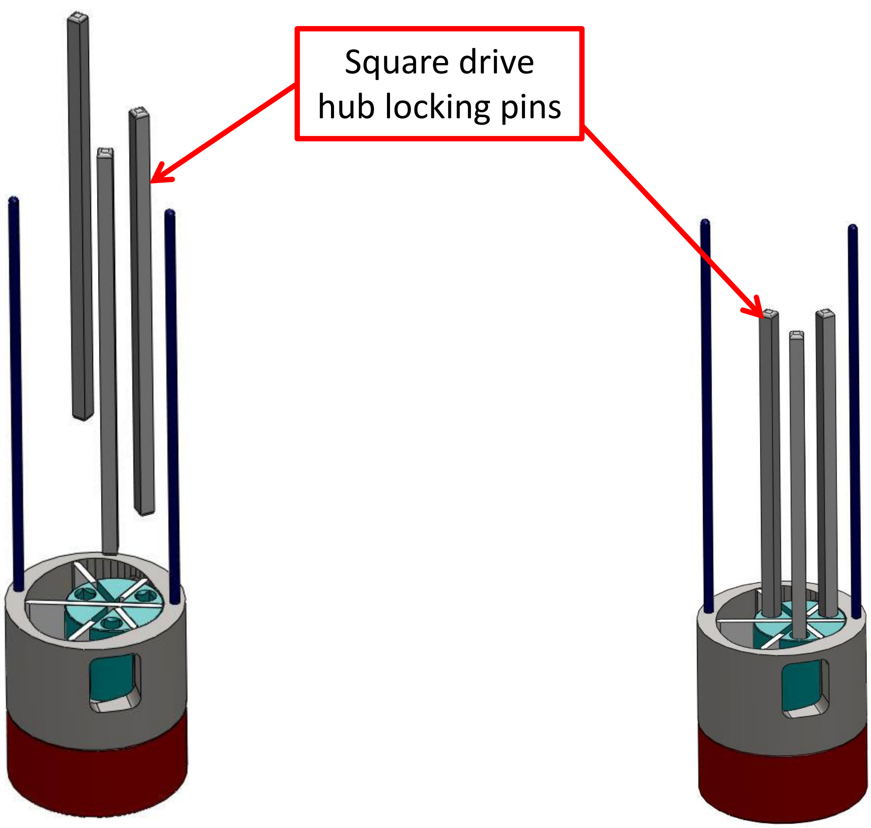
Install first

All vanes installed



3. WATERWHEEL SETUP

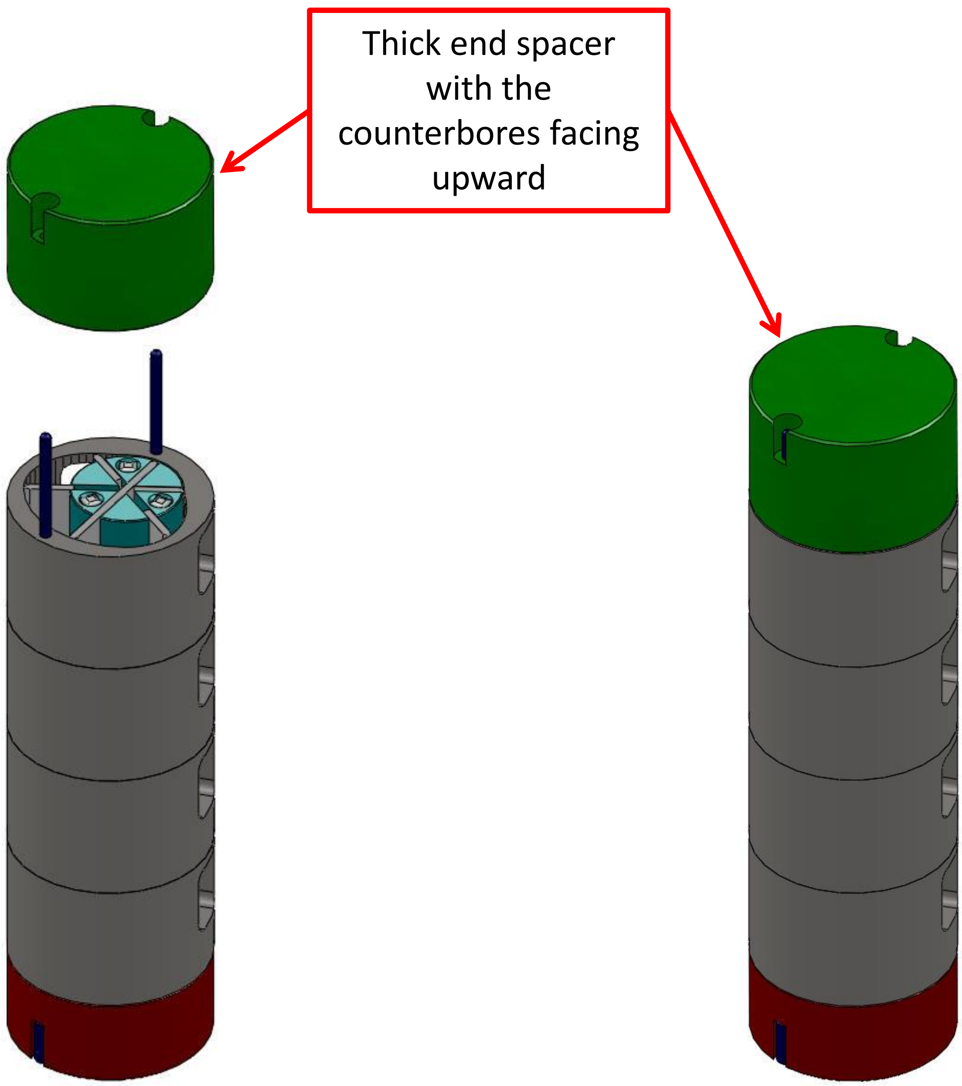
5. Install the square hub locking pins into the square holes in the drive hub. Three (3) pins are required.



6. Repeat steps 2 thru 4 to install the remaining chamber assemblies. Once installed, ensure that the chambers fit together with no gaps in between. Turn the drive hubs and vanes together using the three square locking pins. Ensure that there is no binding between the vanes and the inner walls of the chambers.

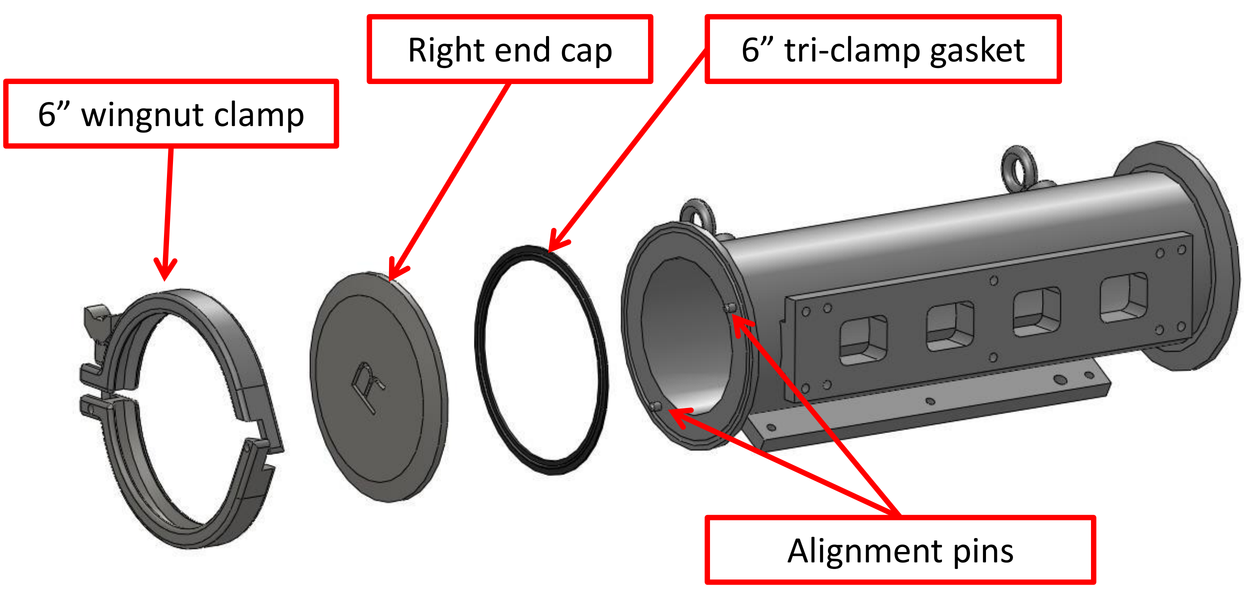
3. WATERWHEEL SETUP

7. Slide the **THICK** end spacer over the chamber locking pins, onto the last chamber. Ensure that the thick end spacer is oriented with the counter bores facing upward, away from the previously assembled chambers.



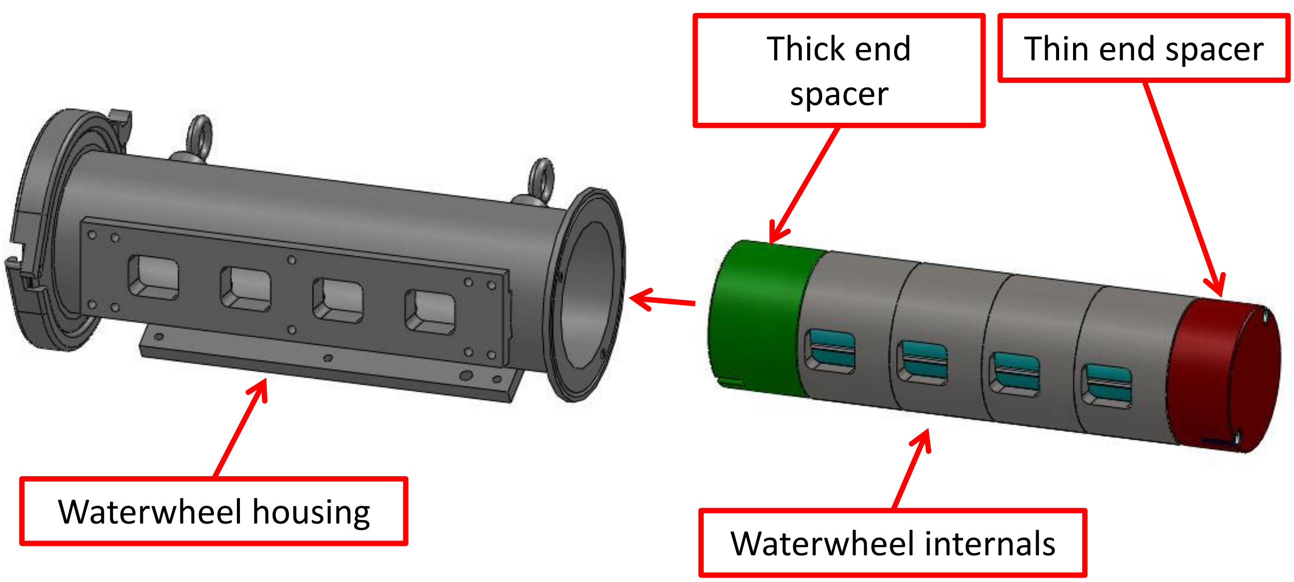
3. WATERWHEEL SETUP

8. The Waterwheel housing is attached to the stand. Place a 6” tri-clamp gasket on the stand-side end of the Waterwheel housing. The gasket will fit onto the groove on the end of the housing. If the stand-side end of the housing has two alignment pins, place the right end cap, labeled with an “R”, onto the end of the housing with the gasket. If the stand-side end of the housing has two alignment holes, place the left end cap, labeled with an “L”, onto the end of the housing with the gasket. The end caps contain holes/pins that will align with the pins/holes on the end of the housing. Clamp the end cap onto the housing using a 6” wingnut clamp.

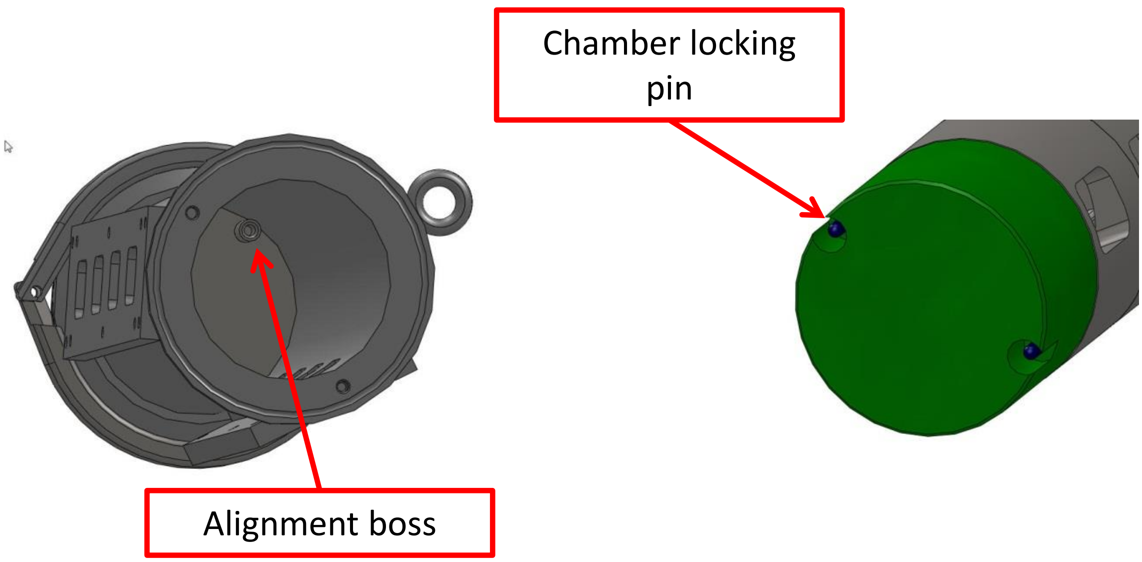


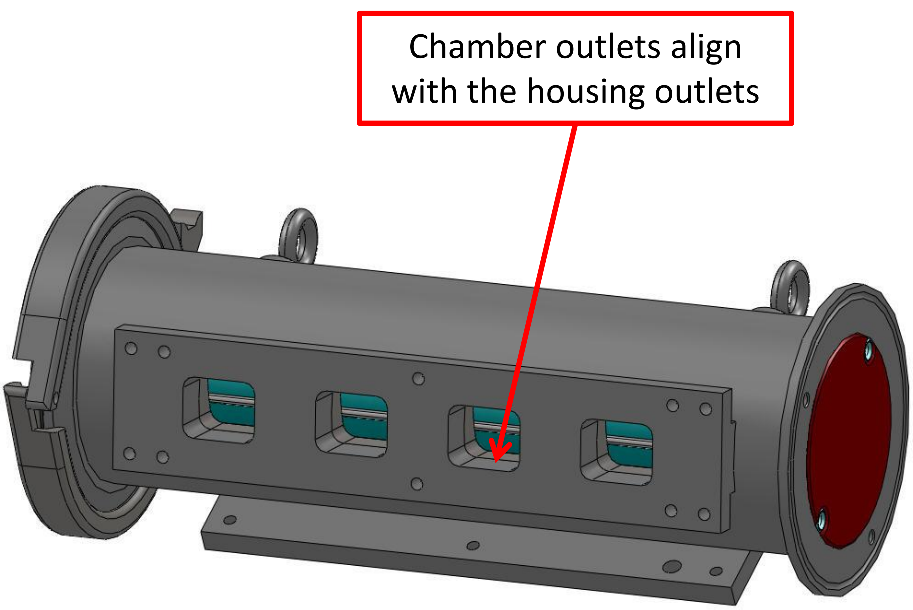
3. WATERWHEEL SETUP

9. Install the Waterwheel internals into the housing. Slide the previously assembled internals into the housing. The **THICK** end spacer must be leading if the **RIGHT** end cap is installed; the **THIN** end must lead if the **LEFT** end cap is installed. Once the internals have been placed into the housing, slowly rotate the internals so that the chamber locking pins in the leading end spacer will engage the pins/bosses that are on the end cap. This will ensure that the outlets on each of the chambers will align with the outlets on the housing. Once the bosses engage, you will be able to slide the internals all the way into the housing.



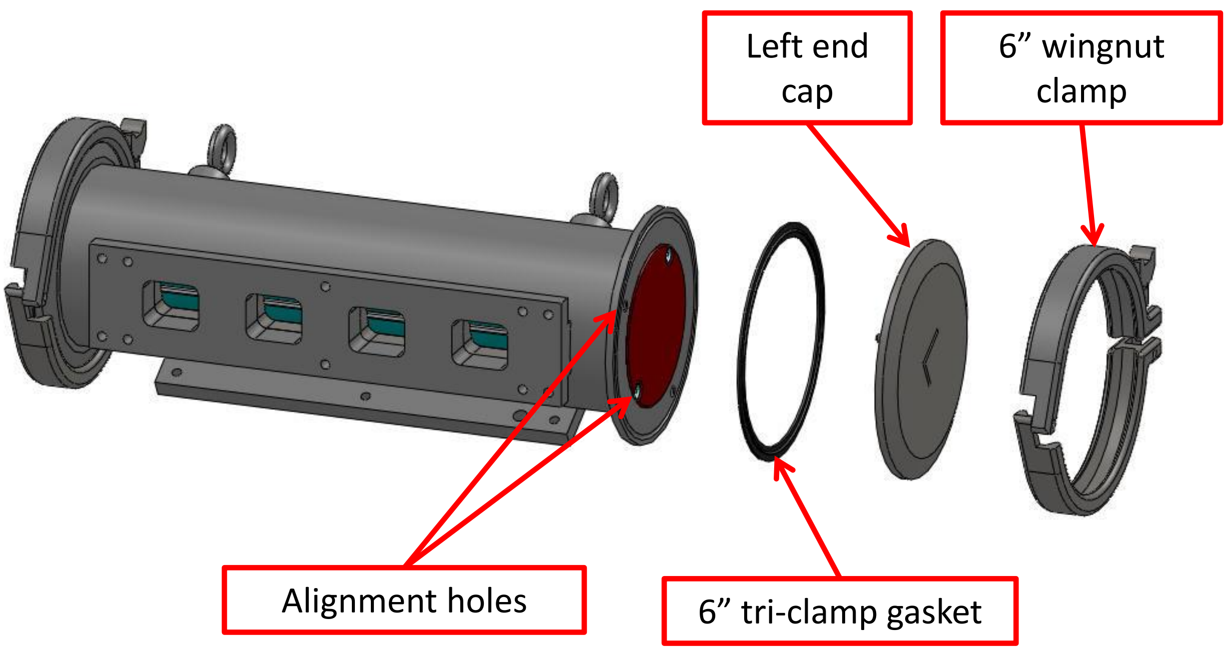
3. WATERWHEEL SETUP





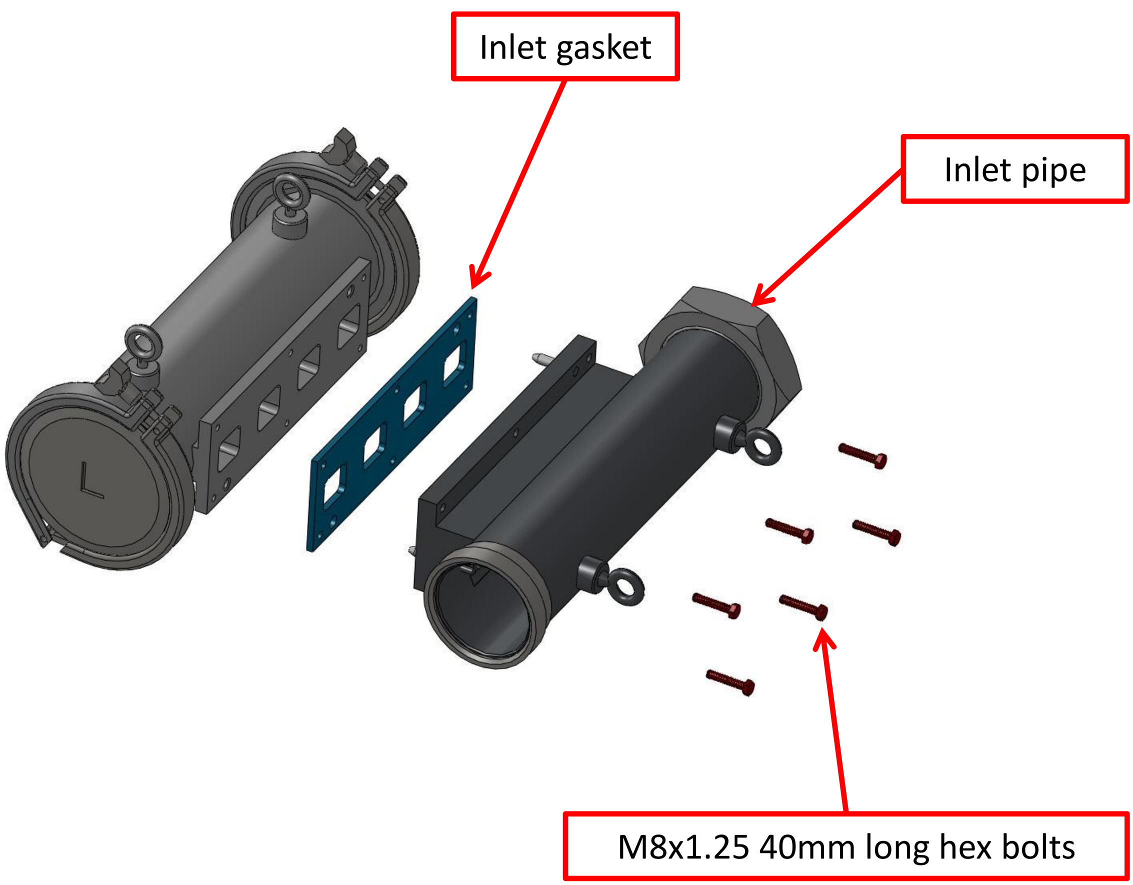
3. WATERWHEEL SETUP

10. Place a 6” tri-clamp gasket on the open end of the Waterwheel housing. The gasket will fit onto the groove on the end of the housing. Place the remaining end cap onto the end of the housing with the gasket. The end cap contains alignment pins/holes that will engage the holes/pins on the end of the housing as well as the showing chamber locking pins. Clamp the end cap onto the housing using a 6” wingnut clamp.



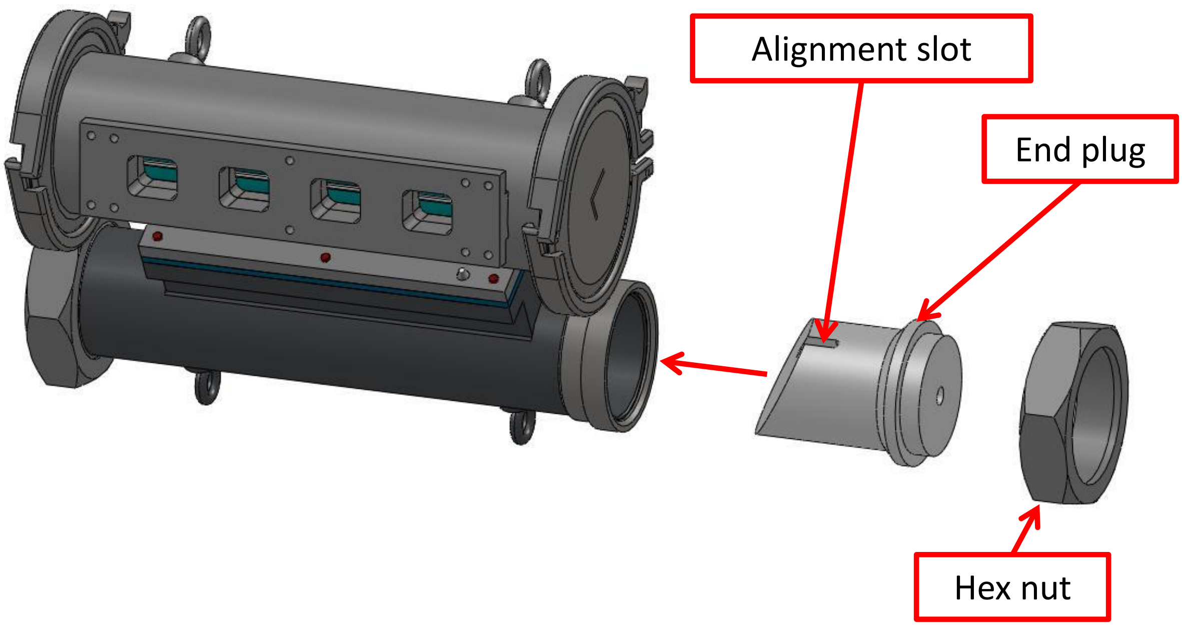
3. WATERWHEEL SETUP

11. Fasten the inlet gasket and the inlet pipe to the housing, using the supplied hex bolts. Apply a food-grade anti-seize lubricant to the threads on each of the bolts prior to assembly. The inlet pipe and inlet gasket are capable of being installed such that the Waterwheel can be supplied with product from either the left or right side.



3. WATERWHEEL SETUP

12. Slide the end plug into the threaded end of the inlet pipe. The inlet pipe contains a pin that will engage the slot in the end plug, ensuring that it is properly aligned. Fasten the end plug onto the inlet pipe using the supplied hex nut.



13. Apply a food-grade anti-seize lubricant to the threads of the inlet pipe and each piece of pipework and connect the pipework from the Vemag filler to the Waterwheel.

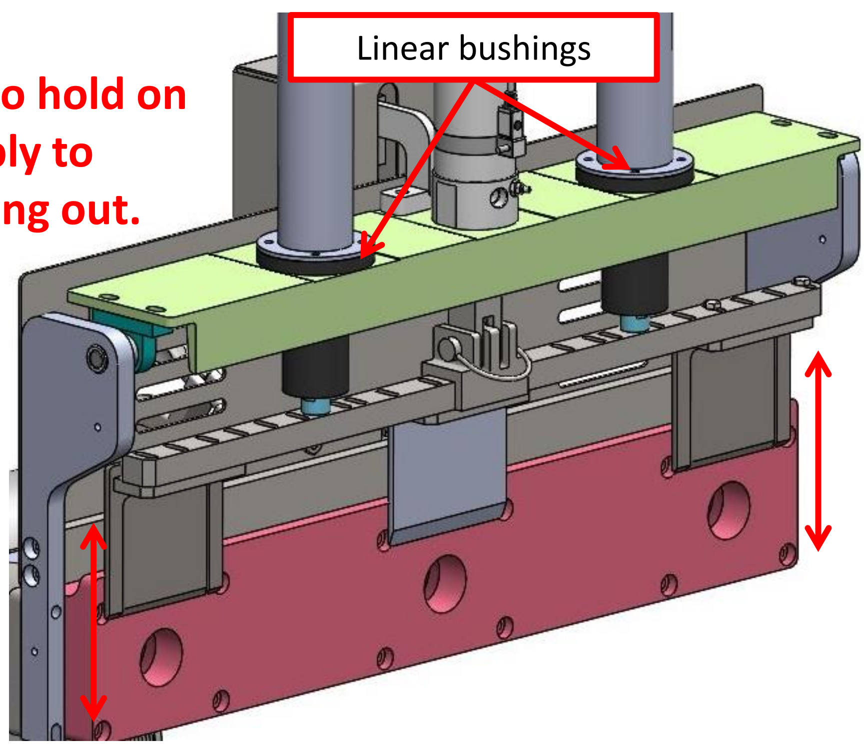
4. CUTOFF SETUP

1. With air line and accessory cable disconnected, make sure all screws are tight. Also, make sure the nut securing the pneumatic knife cylinder is tight.

2. Attach the shaping block and the cutoff to the waterwheel outlet using socket head and/or hex head cap screws. If using a manifold, attach the shaping block and cutoff to the manifold and attach the manifold to the waterwheel outlet.

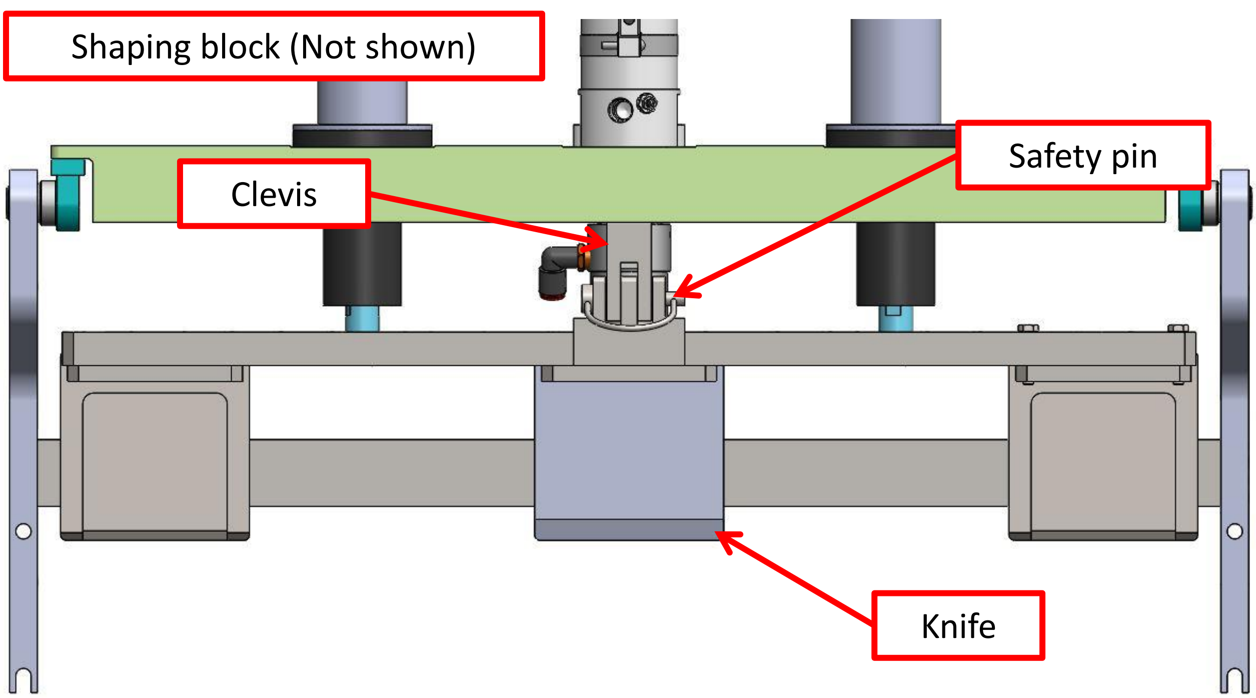
3. Slide the knife guide rods into the bushings and slide it up and down to make sure it slides smoothly without any binding.

CAUTION: Be sure to hold on to the knife assembly to prevent it from falling out.



4. CUTOFF SETUP

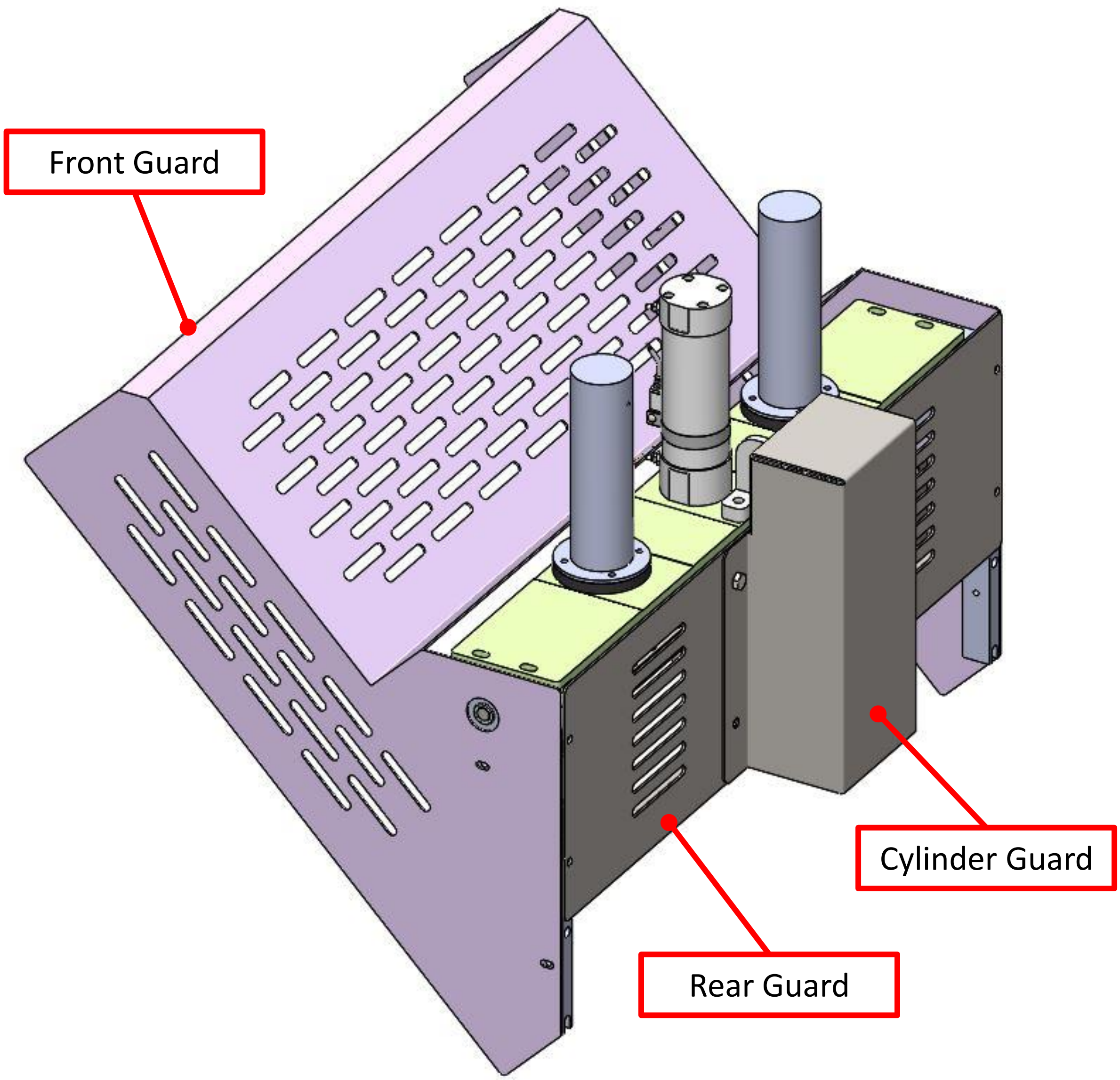
4. While holding on to the knife assembly, connect the knife to the clevis with the safety pin (or connect with the clevis pin and lock it with the hitch pin if applicable). Make sure the knife edge clears the openings in the up position and travels past them in the down position. **The blade(s) must not be able to travel past (off of) the block in either direction.** The knife position can be adjusted by removing the knife from the clevis, loosening the jam nut, and turning the clevis on the pneumatic cylinder shaft. Make sure to re-tighten jam nut after adjusting. Also check that the knife sits flush against the shaping block when pushed against it.



4. CUTOFF SETUP

5. Fasten the front guard to the cutoff body with hex head cap screws.

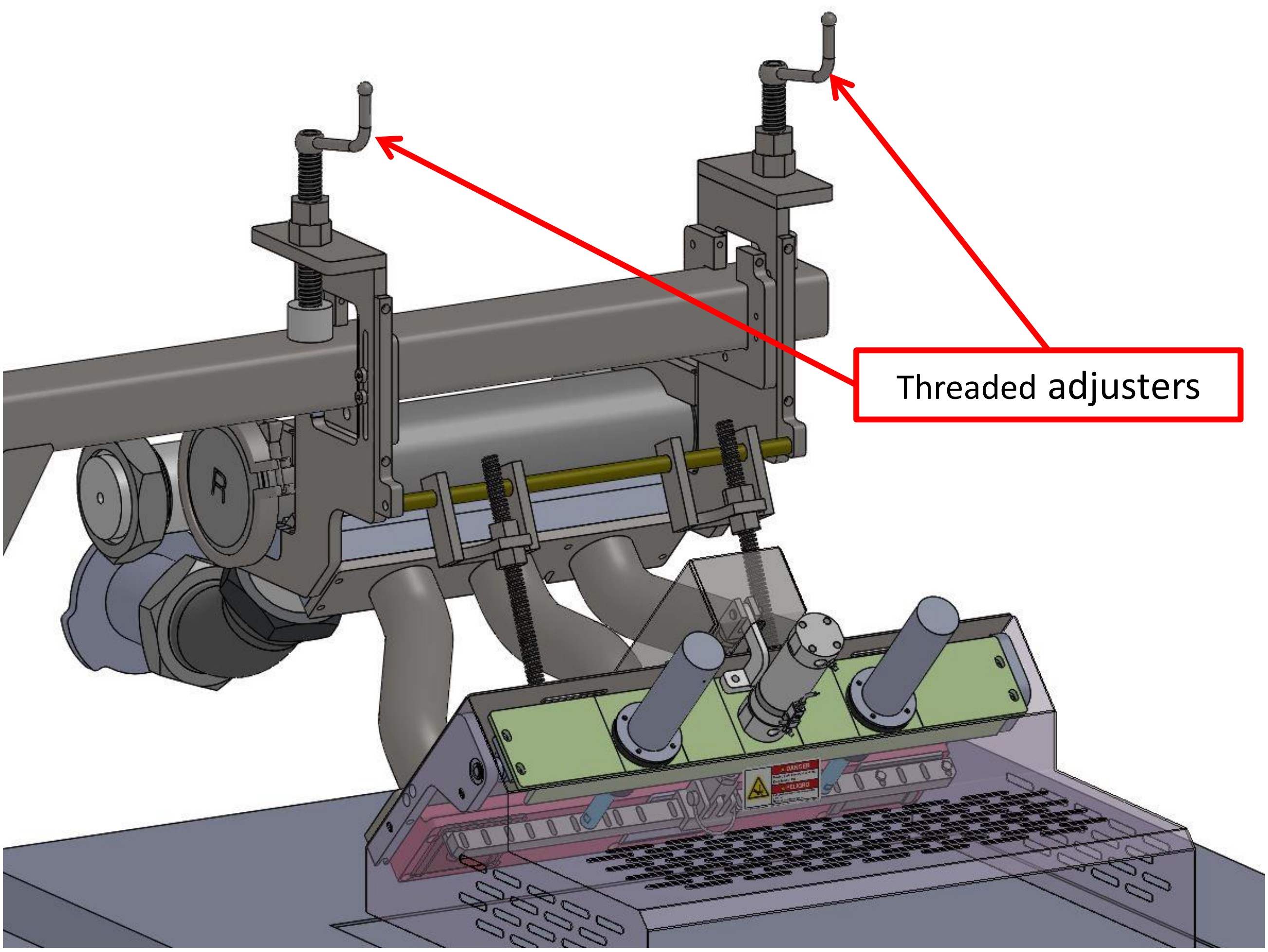
6. Fasten the rear guard to the cutoff with hex head cap screws.



4. CUTOFF SETUP

7. Move the Waterwheel with cutoff into position over the conveyor and bolt the stand to the conveyor through the docking arm (if available). Use the threaded adjusters to move the Waterwheel up and down as necessary.

8. Tighten the threaded adjuster jam nuts after moving the Waterwheel into position.



4. CUTOFF SETUP

9. Connect air lines to the pneumatic cylinders. For prolonged life of the pneumatic cylinder and valve, a user provided filter regulator should be installed directly before the pneumatic cylinder.

AIR COMPRESSOR REQUIREMENTS (80 CUTS/MIN MAX): 80-100 PSI, 5 CFM

10. If a spray bar has been provided, connect water to the bar. A user provided pressure regulator should be used to control the amount and pressure of water sprayed onto the cutoff.

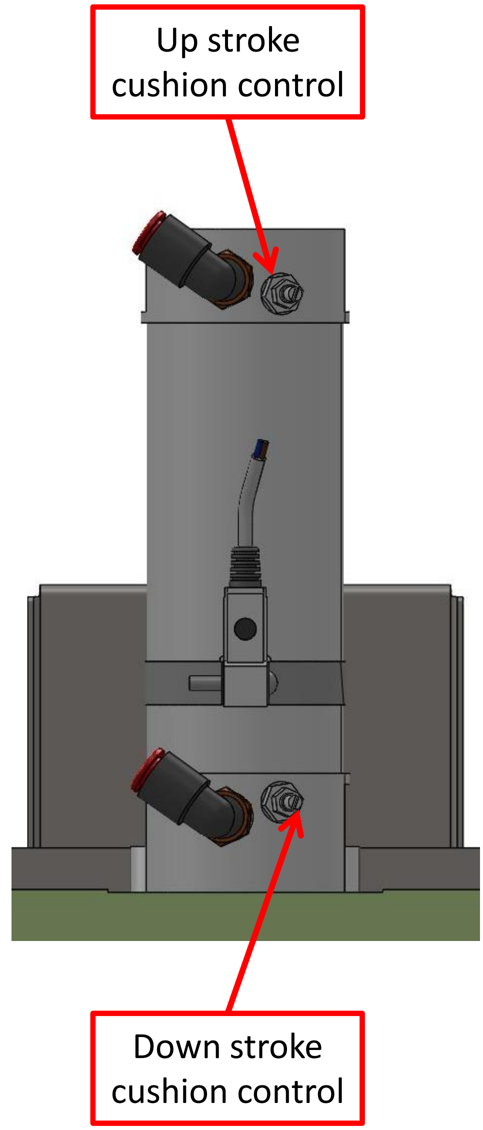
4. CUTOFF SETUP

11. The adjustments on the cylinder can be used to maximize performance of the cylinder and prevent the device from unnecessary wear due to the cylinder bottoming out with great force.

    CAUTION: Do not make adjustments without disconnecting the main air supply first.

    The cushion controls for the cylinder should be adjusted so that the cylinder rod does not slam in either the up or down direction. Adjustment will help to slow the piston rod down when reaching its fully retracted position.

    To make adjustments, loosen the lock nut and turn the adjustment screw. After adjustment, tighten the nut to lock it into position.



4. CUTOFF SETUP

12. Connect the accessory cable and the conveyor proximity switch cable to the Vemag, and turn the filling machine on.

5. OPERATION

The guillotine cutoff is controlled by the Vemag filler. Depending on the product, the cutoff can be operated in several modes. The standard operating mode is normally open. This means the knife is in the up position when the Vemag is not portioning. See section 10 for optional operating modes.

There are three parameters that must be entered into the Vemag program - portion weight in grams, pause (dwell time between portions) in milliseconds, and the clip or twist time in milliseconds. Please refer to the Vemag operating manual for programming instructions.

When the knee lever is pushed, or if a remote start signal is given to the Vemag, the filler will dispense the programmed amount of product then stop and pause. When the portion ends, a 24 vdc signal is sent to the cutoff for the amount of time in milliseconds that is programmed in the clip setting of the PC. This activates the pneumatic cylinder which drives the cutting blade downwards. Once the signal goes away, the cylinder rod and knife blade will retract.

5. OPERATION

These two events happen simultaneously, so it is important that the pause value is set to at least 1.5 times longer than the clip value. (i.e. pause is set at 120ms, when the clip is set to 80ms.) This will allow the knife to retract before the Vemag begins its next cycle. If the knife is not fully retracted when the Vemag begins to portion, the product will push against the blade and could damage the knife or linear bushings.

In this mode, the knife cylinder will be retracted when air is supplied and the coil is **NOT** energized.
CAUTION: If the knife cylinder is extended when air is supplied, disconnect the air, carefully unbolt the cylinder head (do not lose the internal spring), rotate 180° and reattach. Failure to do so may result in cutoff damage.

The accessory cable part numbers to run in this mode are shown in the table below.

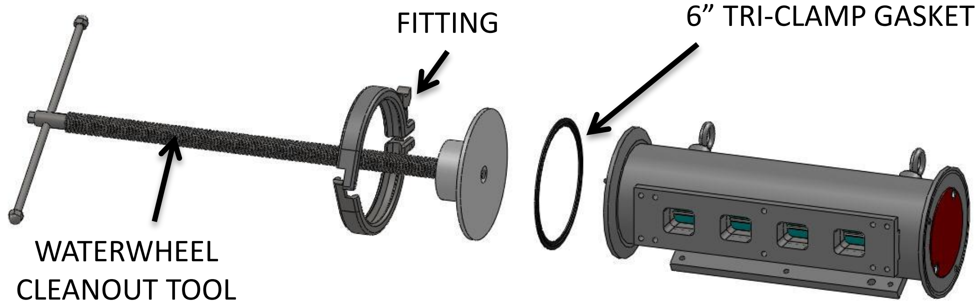
Machine	Cable Part #
V500	CORD-V500
HP, DP, ROBBY, ROBBY II	HPCABLE/HINGED

6. CLEANING

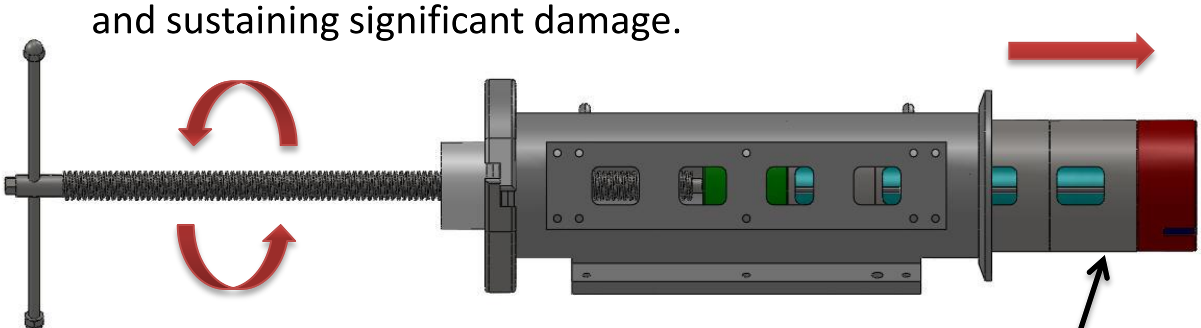
1. **Remove power from the Vemag filler, disconnect the accessory plug and disconnect the air supply from the cutoff. Apply lockout/tagout protocols.**
2. Disconnect all pipework from the Vemag filler to the Waterwheel.
3. Detach the Waterwheel stand from the conveyor dock and roll away from the conveyor. Use the threaded adjusters to move the Waterwheel up and down as necessary.
4. Remove the guarding from the cutoff.
5. Disconnect the knife assembly from the clevis by removing the hitch and clevis pins (or safety pin if applicable).
6. <u>Using caution</u>, slide the knife assembly out of the linear bushings making sure you don't make contact with the cutting edge.
7. Disconnect the shaping block from the Waterwheel outlet.
8. Remove the end plug from the inlet pipe.
9. Remove the inlet pipe from the Waterwheel.
10. Remove the 6" tri-clamp fitting, the end plate, and the 6" tri-clamp gasket from both ends of the Waterwheel housing. This will expose the Waterwheel's internal components on both ends of the housing.

6. CLEANING

11. Attach a 6” tri-clamp gasket to the stand-side end of the waterwheel housing, and attach the cleanout tool to the Waterwheel housing using a 6” tri-clamp fitting. The cleanout tool is used to push the internal components out of the Waterwheel.



12. With the cleanout tool installed, turn the handle on the screw clockwise to advance the screw into the waterwheel. The screw will push the internal components out of the Waterwheel. Have a coworker stand on the opposite side of the Waterwheel to support the internal components as they are pushed out of the housing. This will prevent the internals from falling to the floor and sustaining significant damage.



6. CLEANING

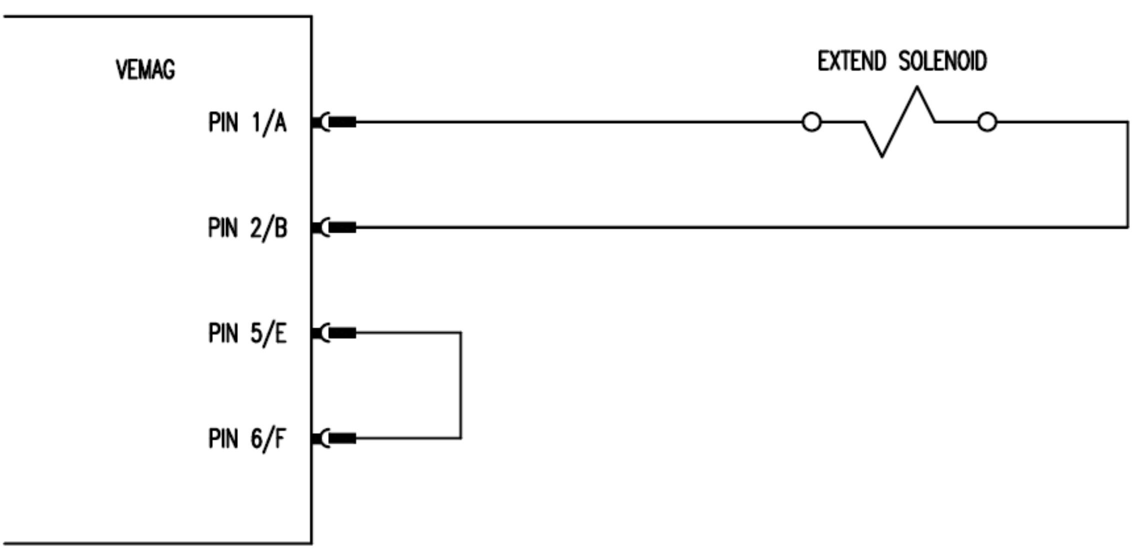
13. Remove the cleanout tool from the housing and disassemble the Waterwheel's internal components.
14. Individual components can now be washed.

7. ELECTRICAL

The Vemag filler will provide a variable length signal at the end of the portion that causes the knife to actuate. The length of the signal is determined by the CLIP time set at the PC. Commonly known as the clip signal, this signal can be advanced or delayed using INFO 8, T-CLIP, on a machine equipped with a PC 880, PC II or PCIII, or the CLIP OFFSET setting on a graphics controller.

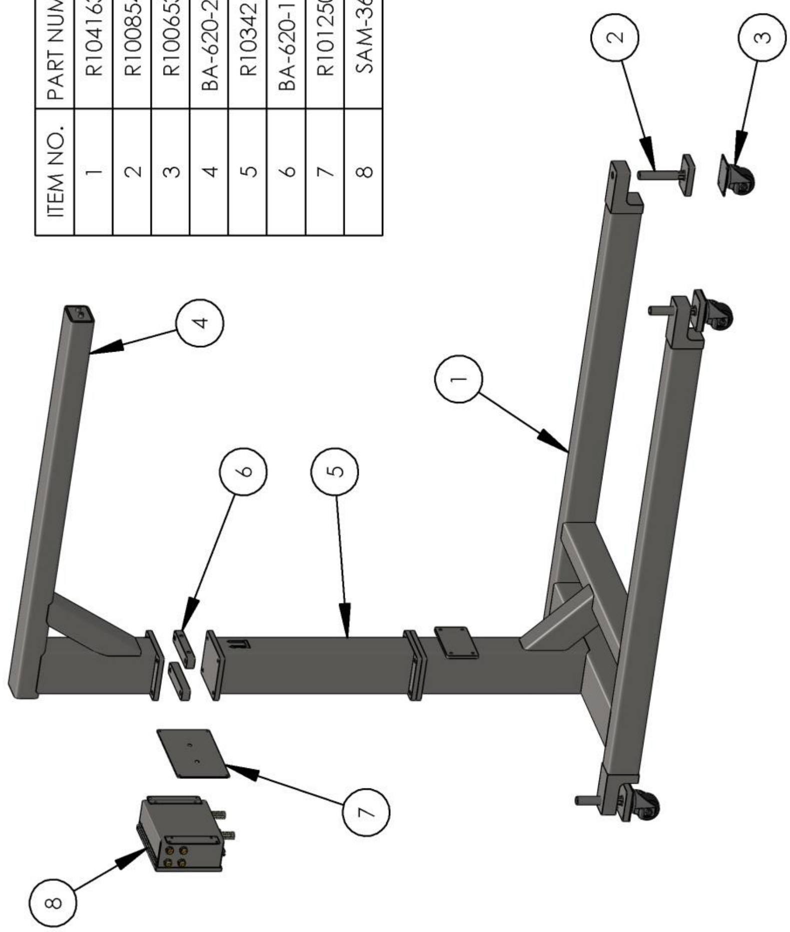
12 PIN CONFIGURATION (V500)

35 PIN CONFIGURATION (HP, DP, ROBBY)



8. PARTS CALL OUT – ASSEMBLY

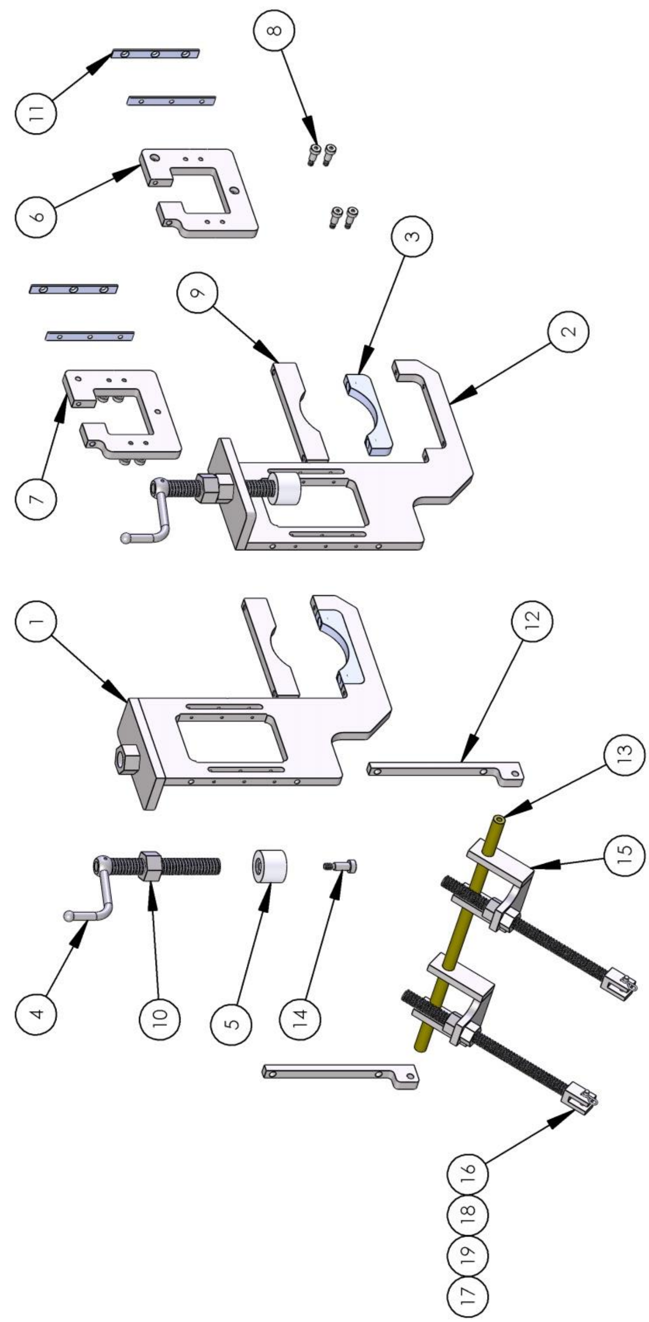
ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	R1041633	WLDMT, MOBILE BASE	1
2	R1008546	CASTER PAD MOUNT	4
3	R1006539	LOW PROFILE SWIVEL CASTER WITH	4
4	BA-620-20-D	STAND BOOM	1
5	R1034214	WLDMT, BOOM HEIGHT EXT. POST	1
6	BA-620-11-A	SPACER, WW STAND, 1"	2
7	R1012502	MOUNTING PLATE,HINGE CONTROL	1
8	SAM-365	CONTROL ENCLOSURE PNEUMATIC	1



8. PARTS CALL OUT – SUPPORT

ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	R1026461	BRACKET, WW CRADLE, REMOTE	1
2	R1026458	BRACKET, WW CRADLE, REMOTE	1
3	BA-620-06-A	PLASTIC CRADLE,WW	2
4	BJ-400-37-A	WW CRANK HANDLE	2
5	BJ-400-38-A	CRANK HANDLE PAD	2
6	R1026997	BRACKET,WW H/A SUPPORT	1
7	R1027015	BRACKET,WW H/A SUPPORT	1
8	MP-080-69	SH. SCREW,M8,10 DIA X18 Lg	8
9	BA-616-39-A	WW RETAINER,TOP	2
10	95066A215	1"-5 ACME HEX NUT	2

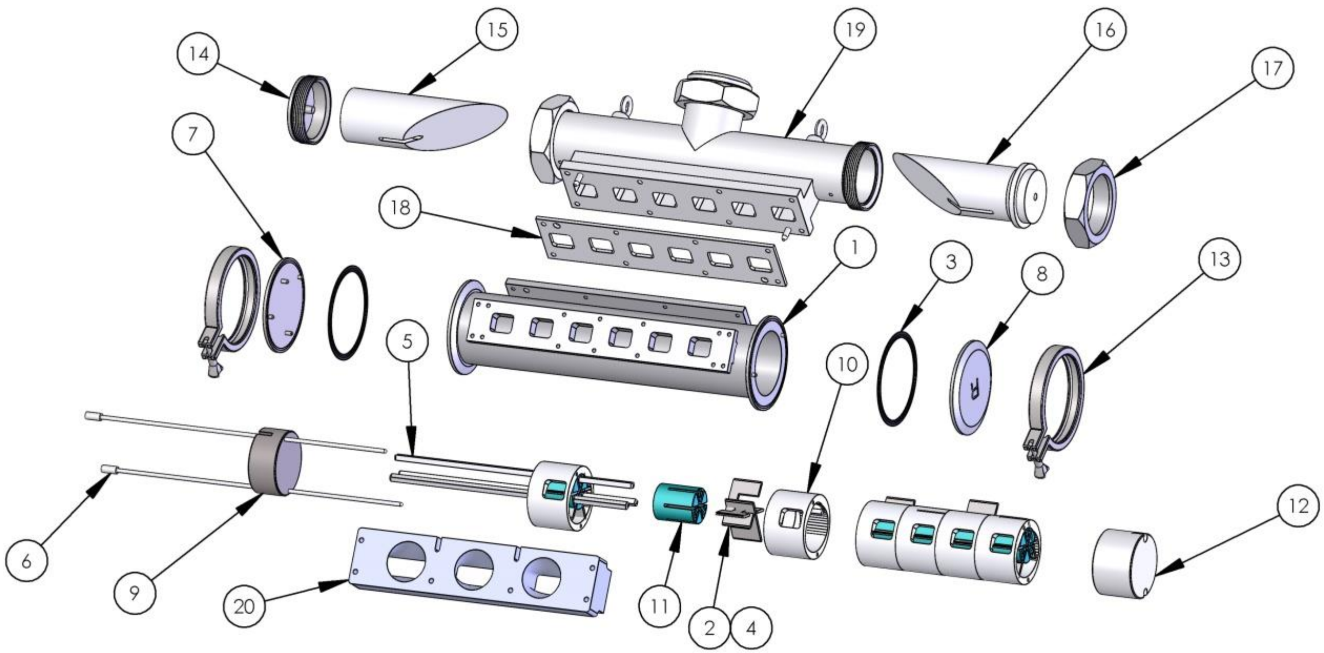
ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
11	R1010526	DELRIN SLIDE, 0.625" X 5.688"	4
12	R1026460	BRACKET, C/O ANGLE PIVOT	2
13	R1039662	BAR, 3/4" DIA. SPACE	1
14	MP-080-46	SH. SCREW, M10, 12 DIA X 25 Lg	2
15	R1026947	BRACKET, C/O CROSS SUPPORT	2
16	98980A142	ROD, C/O ANGLE PIVOT	2
17	R1026942	3/4"-5 ACME NUT	4
18	SSA1545A	CLEVIS PIN;	2
19	SSA1545B	HITCH PIN	2



8. PARTS CALL OUT – WATERWHEEL

ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	BA-342-06-D	WATERWHEEL HOUSING	1
2	BA-337-384-A	VANE, HARDENED TOP/BOTTOM	12
3	40MOU-600	6" SANITARY GASKET	2
4	BA-337-377-A	VANE, HARDENED MIDDLE	6
5	BA-342-199-A	PIN, SQUARE HUB LOCKING	3
6	BA-342-23-A	PIN, CHAMBER LOCKING	2
7	BA-353-17-B	LEFT HAND WW END CAP	1
8	BA-353-16-B	RIGHT HAND WW END CAP	1
9	BA-345-03-A	END SPACER, THIN	1
10	BA-337-365-B	CHAMBER, 4/5 UP, SS HARDENED	6

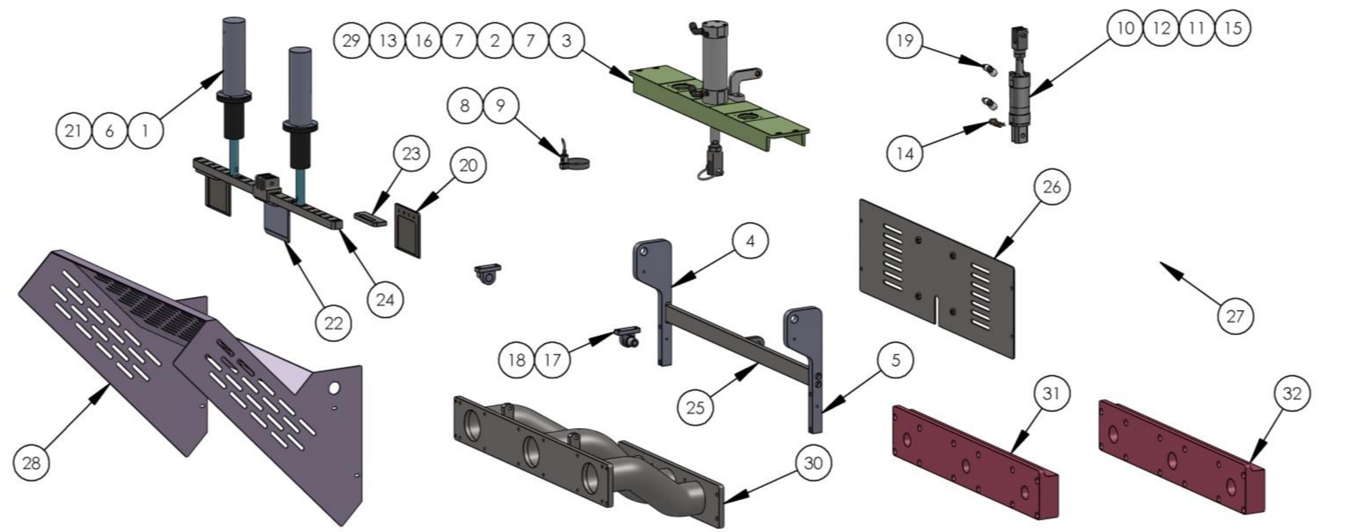
ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
11	BA-337-129-A	DRIVE HUB; STD. CL 3-VANE WW.	6
12	BA-337-199-A	END SPACER, THICK	1
13	P13MHHM-6"	6" SANITARY CLAMP	2
14	BA-486-37-L	SIDE PLUG WELDMENT, 4" INLET	1
15	BR-322-30-B	END PLUG FOR T-STYLE 4" INLET PIPE	1
16	BR-322-41-B	END PLUG, 4" INLET PIPE, REAR FEED	1
17	13H-4	SANITARY NUT	1
18	BA-342-11-B	GASKET; 6- UP INLET	1
19	R1041960	WLDMT, 4" INLET TUBE, 6UPWW	1
20	R1012170	TRANSITION BLOCK,6UPWW TO	1



8. PARTS CALL OUT – CUTOFF

ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	BA-611-14-A	CAP, CYLINDER GUIDE ROD	2
2	R1012182	HINGE DRIVE ARM,CUTOFF,HEIGHT=	1
3	R1038796	CYL MOUNTING PLATE 26" Lg	1
4	R1038801	CYLINDER SUPPORT BRACKET,HINGE	1
5	R1038803	CYLINDER SUPPORT BRACKET,HINGE	1
6	DFB12RR	LINEAR BUSHING EXTENDED BODY	2
7	R1032353	SS CYLINDER, 50MM BORE, 75MM	1
8	R1008841	MOUNTING BRACKET, AUTO SWITCH;	1
9	R1008835	SWITCH, BAND MOUNT; SOLID	1
10	CG5EN40TNSR-19-DUM02297	S.S. CYLINDER,40mm BORE x 19mm STROKE	1
11	SSA245	ROD CLEVIS, HINGE CYLINDER	1
12	SSA-226	5/8"-11 JAM NUT	1
13	98480A015	RETAINING PIN,5/16" DIA. x 1-3/8"	1
14	92390A896	CLEVIS PIN WITH HITCH;	1
15	SSA1545A	CLEVIS PIN;	1
16	R1019711	CLEVIS, ROD, M18X1.5, 5/16 PIN	1

ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
17	BA-481-23-A_2017	ASSY, BUSHING & BRKT	2
18	57785K95	UHMW FLANGED SLEEVE BEARING	2
19	NAS2201F-N01-09-S	1/8 NPT x 5/16 FLOW CONTROL	2
20	R1041562	KNIFE, 4.25" W X 4.25" H	3
21	R1011720	GUIDE ROD, 3/4" DIA. 11" LONG	2
22	R1041568	KNIFE, 4.25" W X 4.25" H	3
23	R1012270	ADJUSTMENT PLATE,KNIFE,	3
24	R1038793	KNIFE MOUNTING BAR, 25-1/2" Lg	1
25	R1038745	TIE BAR, HINGE CUTOFF	1
26	R1039769	REAR GUARD,HINGE CUTOFF	1
27	R1040017	GUARD, 3/4" STROKE CYL	1
28	R1040251	FRONT GUARD,HINGE CUTOFF,	1
29	MP-020-3	ELBOW FITTING, 1/4" NPT MALE	2
30	R1038736	WLDMT, 3UP MANIFOLD, 3" DIA	1
31	R1040936	SHAPING BLOCK, 3UP	1
32	R1038744	SHAPING BLOCK, 3UP	1



9. RECOMMENDED SPARE PARTS

QTY	DESCRIPTION	PART NUMBER
3	SAFETY LABEL	MP-010-001
3	CLEVIS PIN	SSA1545A
3	HITCH PIN	SSA1545B
2	6” SANITARY GASKET	40MOU-600
2	PIN, CHAMBER LOCKING, 3UP WW	BA-340-22-A
1	END SPACER, THIN	BA-345-03-A
6	STAINLESS STEEL CHAMBER	BA-337-365-B
3	PIN, SQUARE HUB LOCKING, 6UP WW	BA-342-199-A
6	DRIVE HUB	BA-337-129-A
12	VANE, HARDENED TOP/BOTTOM	BA-337-384-A
6	VANE, HARDENED MIDDLE	BA-337-377-A
1	END SPACER, THICK	BA-337-199-A
1	KNIFE CYLINDER, 2” BORE X 3” STROKE	R1032353
1	HINGE CYLINDER ¾” STOKE	CG5EN40TNSR
2	LINEAR BUSHING	DFB12RR
3	TEFLON COATED KNIFE	R1041562

10. OPTIONAL OPERATING MODES

NORMALLY CLOSED

This operating mode can be used if the product drips during the pause between portions. When the Vemag is not portioning, the knife is in the down position blocking the outlet. While this does not create a perfect seal, it does help prevent dripping for some products.

In this mode, there are two parameters that must be entered into the Vemag program - portion weight in grams, and pause (dwell time between portions) in milliseconds. Please refer to the Vemag operating manual for programming instructions.

When the knee lever is pushed, or if a remote start signal is given to the Vemag, a 24 vdc signal is sent to the solenoid valve causing the knife to raise. When the portion is complete, the voltage is removed, and the knife moves back to the down position.

10. OPTIONAL OPERATING MODES

To operate in this mode, a special cable is required and the actuator on the solenoid valve is set to operate in reverse. Also, if there is little pipe work between the Vemag and the cutoff, it is helpful to give a small amount of time for the blade to raise and get out of the way of the opening before the Vemag begins to portion. On a HP, DP, and ROBBY machines with a PCII, PCIII this delay is set by the T-PORT parameter. On a HP and DP machines with a graphics controller, this it set by the PORTION DELAY parameter. On a V500 this is set on the on delay relay inside the machine, if so equipped (Reiser P/N SAA-341). 50ms is a good number to start with.

In this mode, the knife cylinder will be extended when air is supplied and the coil is **NOT** energized.
CAUTION: If the knife cylinder is retracted when air is supplied, disconnect the air, carefully unbolt the cylinder head (do not lose the internal spring), rotate 180° and reattach. Failure to do so may result in cutoff damage.

Machine	Cable Part #
V500	CORD-V500-DRIPLESS
HP, DP, ROBBY, ROBBY II	HPCABLE/HINGED

10. OPTIONAL OPERATING MODES

UP/DOWN (BREAK)

This operating mode can be used to cut on both the up and down stroke of the knife. This mode is typically used with a wire knife for cutting cheese blocks or similar products.

There are two parameters that must be entered into the Vemag program - portion weight in grams, and pause (dwell time between portions) in milliseconds. Please refer to the Vemag operating manual for programming instructions.

When the knee lever is pushed, or if a remote start signal is given to the Vemag, the filler will dispense the programmed amount of product then stop and pause. When the portion ends, a 24 vdc signal is sent to the cutoff. This activates the pneumatic cylinder which drives the cutting blade downwards cutting the product.

The knife remains in the down position, while the Vemag makes the next portion. After the portion is made, the voltage to the solenoid valve is removed and the knife moves upwards cutting the next portion.

10. OPTIONAL OPERATING MODES

This mode requires a special cable (Reiser P/N HPCABLE-UP/DOWN) for an HP, DP, Robby or Robby II. For a V500, a bi-stable relay kit (Reiser P/N SAA-325) must be installed in the machine.